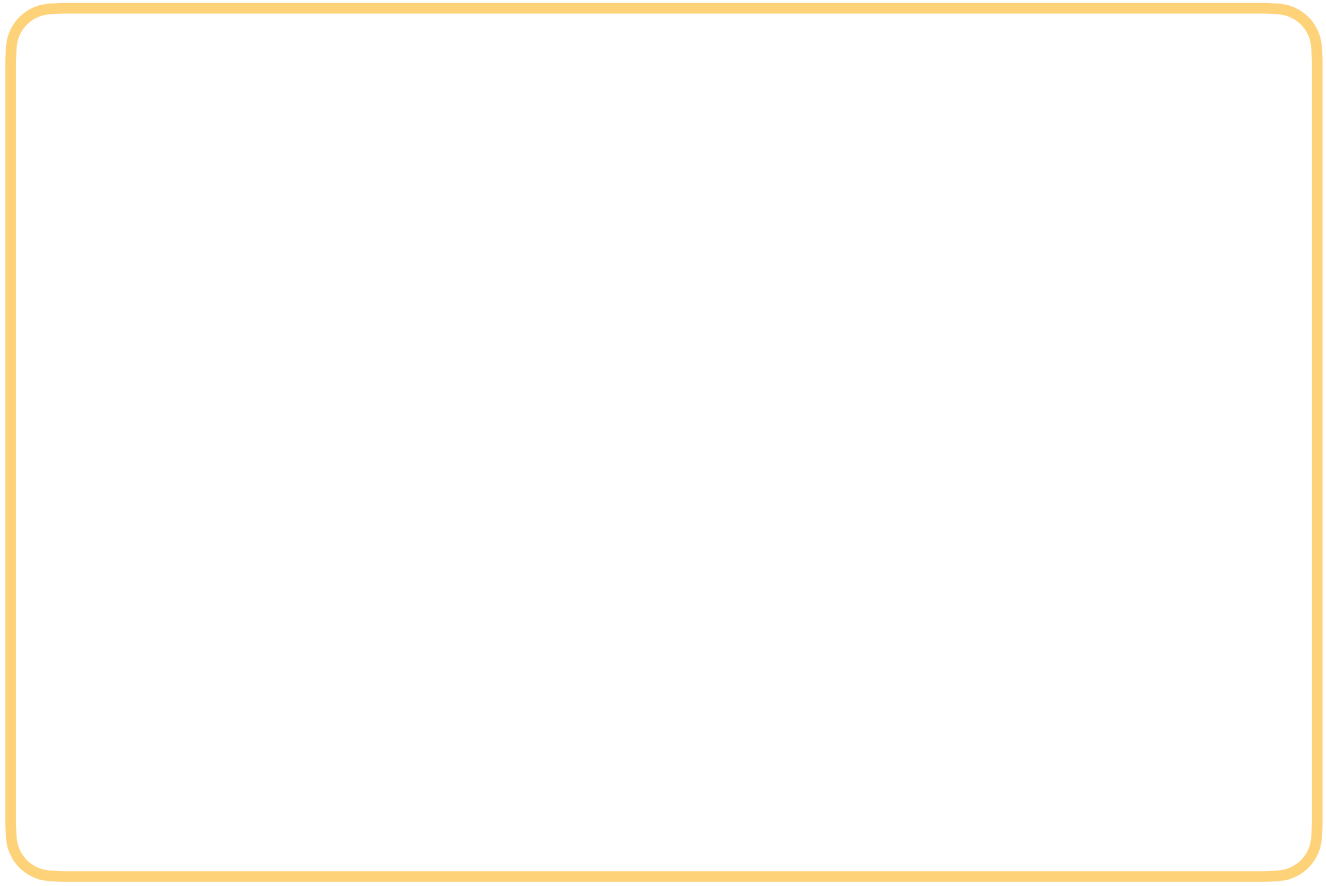
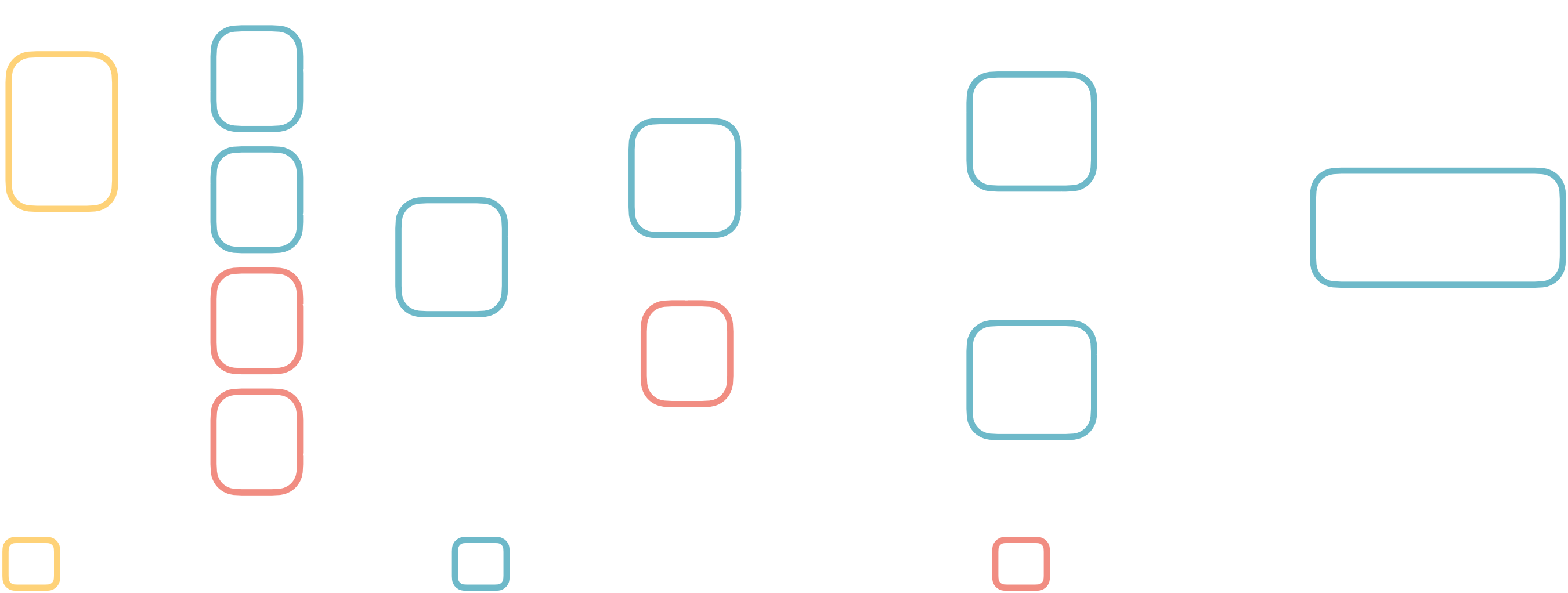


differentiable computation graph



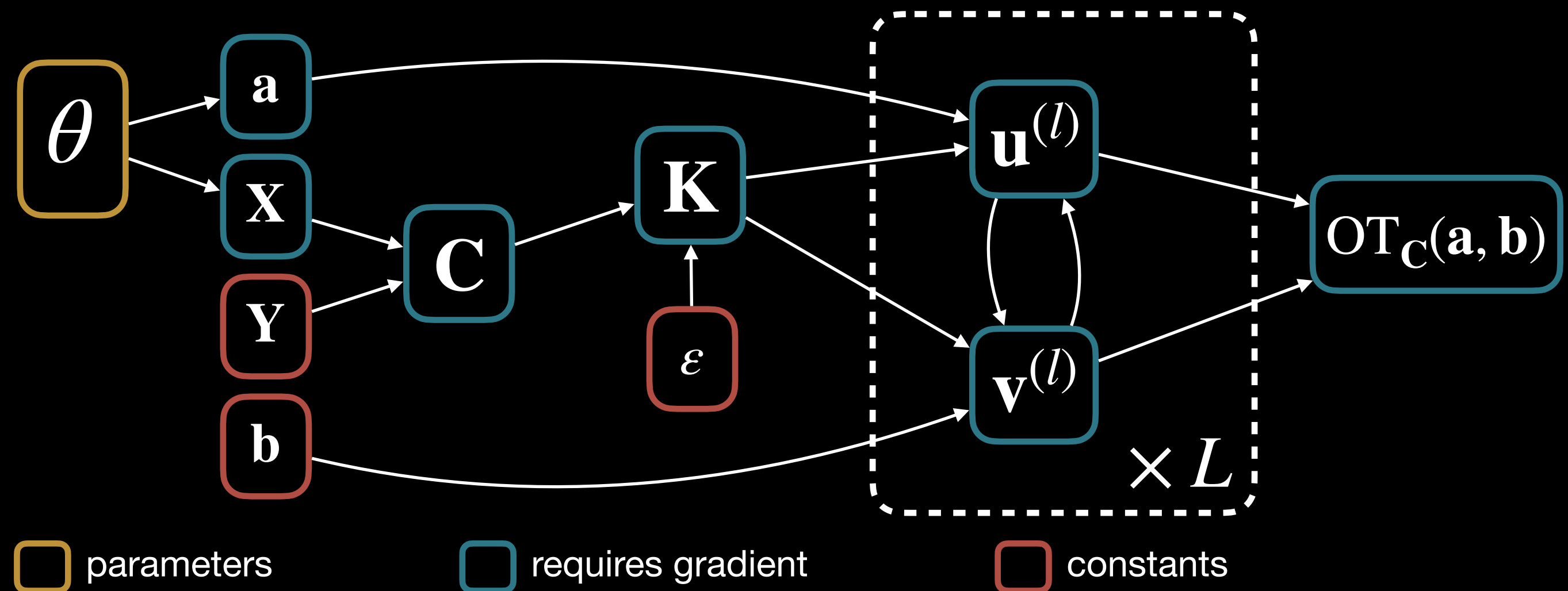


EOT Derivatives via Autodiff

- View the Sinkhorn algorithm as a **differentiable computation graph**
 - Inputs: histograms $\mathbf{a}$, $\mathbf{b}$, support points $\mathbf{X}$, $\mathbf{Y}$ (or cost matrix $\mathbf{C}$).
 - Outputs: regularized OT value, coupling $\pi = \text{diag}(\mathbf{u}^{(L)}) \mathbf{K} \text{diag}(\mathbf{v}^{(L)})$

Sinkhorn Algorithm

1. $\mathbf{K} \leftarrow \exp\{-\mathbf{C}/\varepsilon\}$
2. For l in $\{0, \dots, L-1\}$:
 - $\mathbf{u}^{(l+1)} \leftarrow \mathbf{a} \oslash \mathbf{K} \mathbf{v}^{(l)}$
 - $\mathbf{v}^{(l+1)} \leftarrow \mathbf{b} \oslash \mathbf{K}^\top \mathbf{u}^{(l+1)}$



- Autodiff computes gradients w.r.t. parameters (e.g., θ in $a(\theta)$)
 - Forward: Run L iterations of Sinkhorn (or scaling updates)
 - Backward: Autodiff propagates gradients through all steps

Autodiff: advantages